Calvin Li

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University of California, Davis

2014 - B.S. Computer Science
2016 - M.S. Computer Science
Research Project: Random testing of database engines
June - September 2014

- Randomly generate test cases (SQL queries) in order to look for bugs in database engines
- Test for correctness by transforming queries to equivalent ones and comparing
- Can also test correctness by running queries on multiple engines

Programming Languages and Technologies

Professional Experience

Python	DataDog
C#	Google Cloud
Azure	GitLab
Kubernetes	PagerDuty

Proficient

Java	LaTex
Kotlin	

Experience

C	JavaScript
Swift	Helm
C++	

Personal Projects

Quest
github.com/calvin-li/Quest

- I do Android development as a hobby
- Entirely self-taught
- This is an app I made to help me track tasks using Android's notification system
- Source code on Github, also on the Play Store

Zonar Systems, Inc

March 2023 - Present
Senior Software Engineer, Gateway Integrations team (formerly OEM)

- Lead the development of the "Harsh Events" Light Duty feature
  - Harsh events use the on-device accelerometer detects when vehicle accelerates or brakes too quickly
  - Responsible for the process of translating raw data from device into events that can go into our backend database
  - Involves working with both upstream device team and downstream database team to come up with a design that integrates well with both systems
- Lead the investigation into including State of Charge (SoC) information for electric vehicles with a light duty device
  - This data was readily available from the device, but our legacy backend was not designed to handle EV data.
  - There was a new backend in beta that we wanted to publish SoC information to, while simultaneously publishing the same information without SoC to the legacy backend
  - Involved working heavily with our Database and internal API teams to figure out a solution that was low-risk and minimal effort for all teams involved
- Various non-technical leadership tasks throughout the year
  - Helped organize the company-wide hackathon. One of team of about 10, mainly responsible for choosing the prizes
  - Was on the Diversity, Equity, Inclusion committee, helping come up with ideas and organize events to improve DEI across the company
  - Mentored our summer intern. Team wanted to bring him on full time but we had a hiring freeze.
  - Helped our European counterparts integrate into our system and take our Light Duty product to the next level. This included flying them out to Seattle so they can work with us face-to-face.

March 2020 - March 2023
Software Engineer II, OEM team

- Maintained and improved existing OEM services
  - Services generally ingested data from partner hardware, processed, then inserted data into Zonar's systems
  - Required collaborating directly with multiple different partner companies (eg DTNA, Cummins), each with their own hardware and systems

  - Learned the systems well enough to be part of the on-call rotation within 3 months of joining
  - Improved DevOps and helped uplift the services to the cloud
- Helped develop Light Duty TCU (Telematics Control Unit) system
  - A simpler, cheaper version of Zonar's existing product geared towards light vehicles (eg cars and pickups) instead of large trucks
  - Collaboration with parent company (Continental Ag), who developed the hardware
  - System developed from the ground up as a part of a larger OEM platform, designed to eventually replace all the various services we currently have
  - Saw the product through from planning to launch (Dec. 2021) and beyond
  - Developed a load testing system that takes previously recorded input and replays it to simulate a system under load
- Helped plan and develop the OEM Platform
  - This is a restructure of our services to be more generalized and use more shared code
  - Beginning with Light Duty TCU, and then integrating out older services
  - The idea is to create a common framework to ingest data from various sources, process it, and then send it to our backend
- OEM systems made up of micro-services, mix of Google Cloud Functions and containers in GKE. Data handled using MemoryStore (Google Redis) and Google DataStore.
  - Python used throughout
  - Heavy use of GitLab CI for continuous pre-prod deployments, and quick prod releases
  - Use Terraform to manage Google Cloud Infrastructure through code
- They're based in Seattle, but have been primarily remote since I joined

Zume, Inc

November 2019 - March 2020
Software Engineer

- Worked on web app for tracking production at packaging factory
  - Improved CI/CD of system, bringing it in line with company wide standards
  - Learned some web development and containers

Microsoft Corporation

August 2017 - June 2019
Software Engineer, Microsoft Graph

- Proxy API for AAD Graph and other Microsoft APIs
- Split time between feature development and maintenance of 4 nines uptime
- Expanded service into new cloud environment (US government)
- Took ownership of our integration testing service
- Part of the on-call rotation
  - Responded to incidents from automated monitoring or users 24/7
  - Used experience to analyze health of service and make improvements to both project setup and livesite setup
  - Designed and implement improvements to our alerting and troubleshooting.

June 2016 - August 2017
Software Engineer, Azure AD Graph

- Rest API for Azure Active Directory
- Added publisher domain feature to Applications API
  - Lets application owners specify their a domain to show users when asking for permissions
  - Responsible for implementation and test design
- Maintained the product's C# library
  - Helps developer make requests to the API without having to craft HTTP Rest requests
  - Inherited sole ownership upon joining team
  - Fixed bugs, cleaned up the code, and added automation
  - Worked with users to transition them to successor product (Microsoft Graph)

June 2015 - September 2015
Software Engineer Intern, Microsoft Authenticator App

- iOS app for Microsoft's MFA service
- Added usability features (fingerprint authentication and Apple Watch integration)
- Main app coded in Objective-C, watch app coded in Swift